

Practical No. 10

Aim

Mini Project: Design and implement an end-to-end data engineering solution including data ingestion, cleaning, transformation, storage, data warehouse design, and reporting.

1. Data Ingestion

```
All datasets loaded successfully!
Products: (77, 10)
Orders: (830, 14)
Order Details: (2155, 5)
```

2. Data Cleaning

```
# Check missing values
print("Missing Values:")
print("Products:\n", products.isnull().sum())
print("\nOrders:\n", orders.isnull().sum())
print("\nOrder Details:\n", order_details.isnull().sum())

# Check duplicate records
print("\nDuplicate Records:")
print("Products:", products.duplicated().sum())
print("Orders:", orders.duplicated().sum())
print("Order Details:", order_details.duplicated().sum())

... Missing Values:
Products:
ProductID      0
ProductName     0
SupplierID     0
CategoryID     0
QuantityPerUnit 0
UnitPrice      0
UnitsInStock   0
UnitsOnOrder   0
ReorderLevel   0
Discontinued   0
dtype: int64

Orders:
OrderID      0
CustomerID   0
EmployeeID   0
OrderDate    0
RequiredDate 0
ShippedDate  21
ShipVia       0
Freight       0
ShipName     0
ShipAddress  19
ShipCity     19
ShipRegion   507
ShipPostalCode 19
ShipCountry   0
dtype: int64

Order Details:
OrderID      0
ProductID    0
UnitPrice    0
Quantity     0
Discount     0
dtype: int64

Duplicate Records:
Products: 0
Orders: 0
Order Details: 0
```

3. Data Transformation

All three datasets merged successfully!
Rows and columns: (2155, 27)

OrderID	ProductID	UnitPrice_x	Quantity	Discount	CustomerID	EmployeeID	OrderDate	RequiredDate	ShippedDate	...	ShipCountry	ProductName	SupplierID	CategoryID	QuantityPerUnit	UnitPrice_y	UnitsInStock
0	10248	11	14.0	12	0	VINET	5	1996-07-04	1996-08-01 00:00:00	1996-07-16 00:00:00	...	France	Queso Cabrales	5	4	1 kg pkg.	21.00
1	10248	42	9.8	10	0	VINET	5	1996-07-04	1996-08-01 00:00:00	1996-07-16 00:00:00	...	France	Singaporean Hokkien Fried Mee	20	5	32 - 1 kg pkgs.	14.00
2	10248	72	34.8	5	0	VINET	5	1996-07-04	1996-08-01 00:00:00	1996-07-16 00:00:00	...	France	Mozzarella di Giovanni	14	4	24 - 200 g pkgs.	34.80
3	10249	14	18.6	9	0	TOMSP	6	1996-07-05	1996-08-16 00:00:00	1996-07-10 00:00:00	...	Germany	Tofu	6	7	40 - 100 g pkgs.	23.25
4	10249	51	42.4	40	0	TOMSP	6	1996-07-05	1996-08-16 00:00:00	1996-07-10 00:00:00	...	Germany	Manjimp Dried Apples	24	7	50 - 300 g pkgs.	53.00

5 rows x 27 columns

Sales Amount calculated successfully!

OrderID	ProductID	Quantity	UnitPrice_x	Discount	SalesAmount
0	10248	11	14.0	0	168.0
1	10248	42	9.8	0	98.0
2	10248	72	34.8	0	174.0
3	10249	14	18.6	0	167.4
4	10249	51	42.4	0	1696.0

4. Data Storage

```
print("File: transformed_sales_data.csv")

Transformed data stored successfully!
File: transformed_sales_data.csv
```

5. Data Warehouse Design

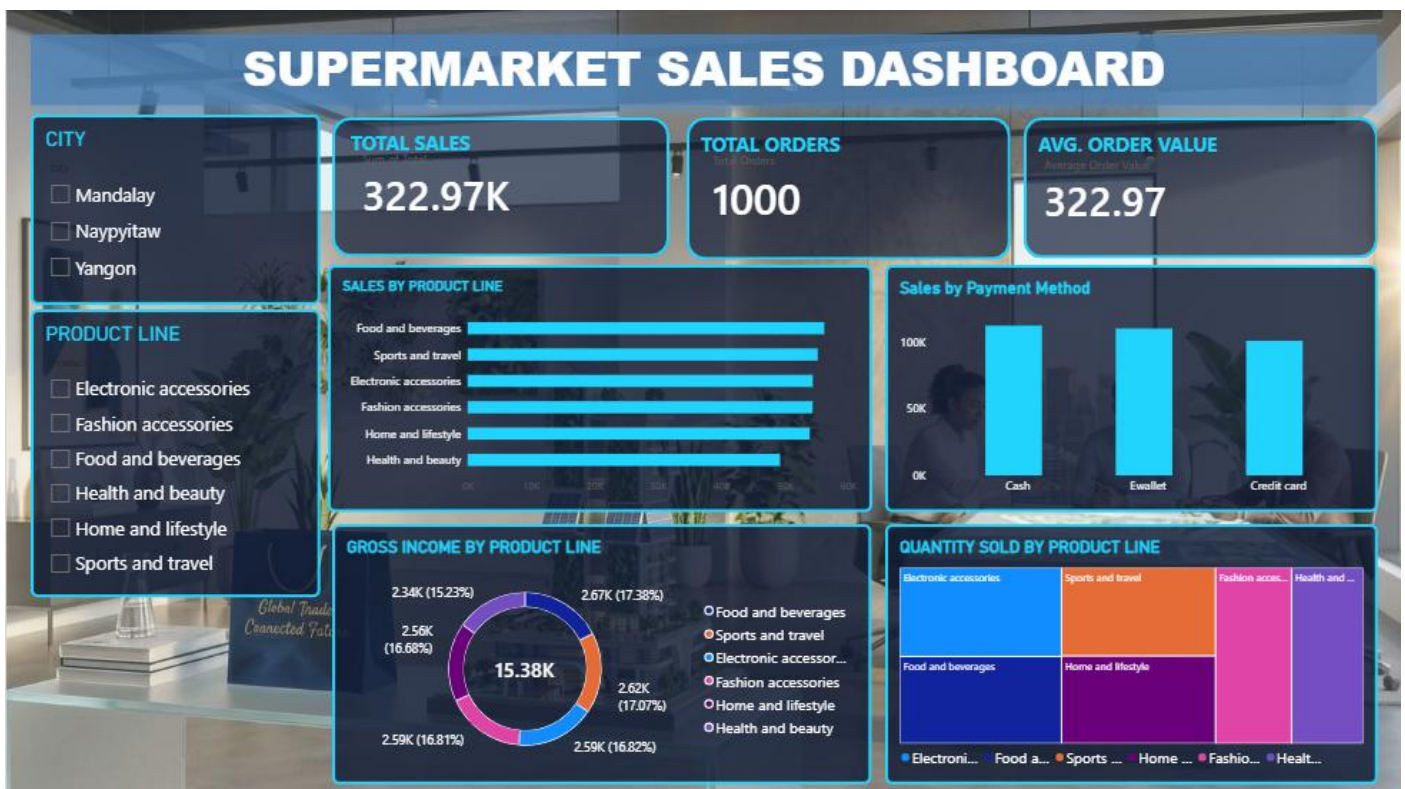
Dimension tables created successfully!
Dim Product: (77, 6)
Dim Order: (830, 6)

FactSales table created successfully!
Rows and Columns: (2155, 6)

	OrderID	ProductID	Quantity	UnitPrice_x	Discount	SalesAmount
0	10248	11	12	14.0	0	168.0
1	10248	42	10	9.8	0	98.0
2	10248	72	5	34.8	0	174.0
3	10249	14	9	18.6	0	167.4
4	10249	51	40	42.4	0	1696.0

*** Data Warehouse tables stored successfully!
DimProduct.csv
DimOrder.csv
FactSales.csv

6. Reporting / Visualization



7. Conclusion

The project successfully implemented an end-to-end data engineering pipeline using multiple CSV datasets. The data was ingested, cleaned, transformed, integrated, stored, and organized into a star-schema-based data warehouse. The processed data was then used for reporting and visualization through Power BI.